



microring modulators.



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- Performance Metrics
- Process
- What's next



contents



Performance Metrics



Performance Metrics

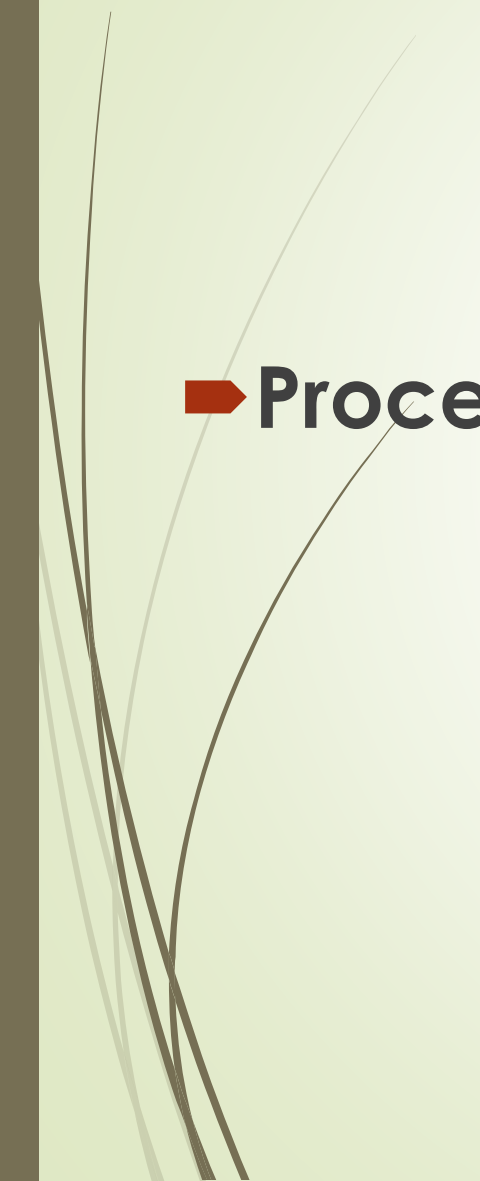
- Insertion loss
- Bandwidth
- Footprint
- Extinction ratio
- FSR
- FWHM
- Q-factor
- Finesse
- $V_{\pi} \cdot L$



contents



Process





Process

- p-n junction diode in silicon.
- plasma dispersion effect.
- Soref equations.
- Resonance.
- Transfer function (proof & matlab).
- Voltage-Phase Relation.
- Voltage – effective ref. index (concept & matlab).
- p-n junction configurations.
- Comparison Table over 9 Silicon Microring Modulator Papers.



contents



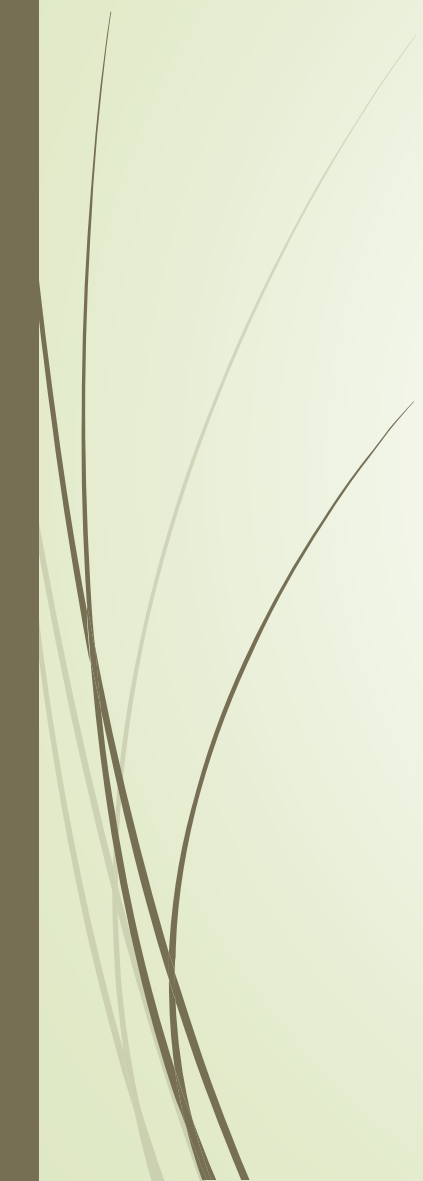
What's next ?





What's next ?

- **Different p–n junction geometry and design trade offs.**
- **Ready to simulate examples on lumerical.**



Thanks 😊